

Abstract

Limited research directly investigates urban sprawl's correlation with widening racial and income inequality. This thesis intends to explore these relationships in the Las Vegas Valley, a racially and economically diverse region with rapidly occurring sprawl. Using sprawl, demographic, and air pollution data from 2010, various quantitative methodologies, including linear regressions and a two-sample t-test, were conducted to test whether areas with greater sprawl are associated with shorter commute times, higher dependence on private vehicles, greater access to higher-paying leadership occupations, and lower levels of air pollution. Additionally, principal component analysis (PCA) was used to create a new "sprawl burden" index that assessed whether people of color and low-income communities disproportionately reside in areas experiencing negative sprawl outcomes. Results found that sprawling areas contain fewer people of color and more high-income residents. These areas are also correlated with longer commute times, greater private vehicle use, greater access to high-paying leadership positions, and less exposure to particulate matter 2.5 (PM_{2.5}). Meanwhile, non-sprawling areas are associated with greater concentrations of people of color and low-income residents. These groups are also more likely to use public transit and other means of transportation, be employed in service-oriented occupations, and face greater PM_{2.5} exposure. Income was found to have the most consistent positive relationship with sprawl, indicating that income may be the strongest predictor of sprawl than any of the other analyzed variables and outcomes. They also suggest that sprawl's impact is more nuanced than what traditional metrics capture, indicating that marginalized narratives need to be considered in the development of sprawl policy.